

REVIEW

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Systematic review of green pilot zones, innovation, and sustainability

Eduardo José da Silva Santana^{1*}, Djalma Silva Guimarães Junior¹ and Eduardo Cardoso Gonçalves¹

*Correspondence:
eduardo.jssantana@upe.br

¹ UPE: Universidade de
Pernambuco, Recife, Brazil

Abstract

Objective: This article aimed to identify the impacts of green pilot zones on innovation and sustainability.

Design/methodology/approach: A Systematic Literature Review (SLR) was conducted in the Web of Science and Scopus databases using the keywords *Innovation*, *Green finance reform*, and *innovation pilot zone*, PZGFRI and GFRI, yielding a final sample of 47 articles.

Results: This study identified key trends through which green pilot zones policy influences green innovation and sustainability, namely geographical location, market competitiveness, development level, resource heterogeneity, environmental regulation, ownership, firm size and pollution intensity, credit access, and digitalization.

Implications: This study provides a comprehensive overview of research on green pilot zones, covering the period from their emergence in 2022 to the most recent studies. It traces the evolution of the topic and offers insights that could help policy-makers and public officials improve existing sustainable policies in specific contexts and/or implement the green pilot zones model in other regions or countries seeking to replicate it.

Originality: Although green pilot zones have recently attracted increasing scholarly attention, this SLR is likely among the first to systematically compile and synthesize the existing body of knowledge on the topic. It identifies the main points of convergence among studies, as well as their divergences and contradictions, thereby advancing a more coherent understanding of the field.

Keywords: Green finance, Green pilot zones, Green innovation, Sustainability



1 Introduction

The rapid economic growth experienced during the twentieth century, largely driven by industrialization, generated substantial wealth for nations [1]. It also led to excessive consumption of fossil fuels and increased greenhouse gas emissions, which have contributed to global climate change and the deterioration of environmental quality [2], resulting in severe environmental problems worldwide [3, 4].

Hou et al. [5] argue that extreme weather events and anthropogenic environmental degradation will constitute the primary global risks in the coming decades. Indeed, the global average temperature increased by 1.09 °C from 2011 to 2020 compared to the period from 1850 to 1900 [2] and it will probably exceed pre-industrial levels by 1.5 °C by 2030 [1].

The consequences of climate change include sea-level rise, reductions in agricultural productivity, deterioration of air quality, freshwater scarcity, and an increased frequency and intensity of extreme weather events. Recent examples include severe droughts in Europe, extreme heatwaves in India, and episodes of heavy rainfall and flooding observed in Brazil (Rio Grande do Sul), Spain (Valencia), the United States of America (Texas), and South Korea [2, 6–10].

In pursuit of high economic growth, China has emerged as one of the principal contributors to climate change in recent decades and one of the largest sources of pollution and greenhouse gas emissions worldwide [5, 11–13].

However, this growth followed a strategy commonly described as "pollute first, clean later", resulting in a strong dependence on energy-intensive industries [13] and leading to substantial increases in resource consumption and ecological degradation [14, 15].

As a result, China became the world's largest emitter of greenhouse gases in 2005 [5, 13] and the largest energy consumer in 2009 [16]. Between 2000 and 2020, total energy consumption increased from 1,469 to 4,983 million tons, corresponding to an average annual growth rate of 6.28% [11]. In 2021, China emitted approximately twice as much carbon as the United States and nearly four times as much as the European Union [17]. Moreover, according to Zhang et al. [13], China's *per capita* carbon emissions are approximately 15% higher than those of advanced economies.

In response, China announced ambitious targets to peak carbon emissions by 2030 and achieve carbon neutrality by 2060 [2, 18, 19].

In this context, China has adopted green finance policies as a central instrument for reducing greenhouse gas emissions while reconciling economic development with environmental protection [5, 20–26]. According to Dong et al. and Huang et al. [27, 28], green finance policies are among the most effective approaches for reducing carbon emissions.

Accordingly, in 2016, the People's Bank of China, together with seven other government departments, issued the "Guidelines for Building a Green Financial System", making China the first country to actively promote the establishment of a comprehensive green financial system [4, 18].

In 2017, Green Finance Reform and Innovation Pilot Zones, hereinafter referred to as green pilot zones (GPZ), were established in five Chinese provinces: Zhejiang, Guangdong, Ghizhou, Jiangxi, and Xinjiang [4, 18, 21, 29]. These pilot zones were designed to advance green finance and foster green innovation [30].

Against this backdrop, a key research question arises: What are the impacts of the GPZ on green innovation and sustainability?

Therefore, this SLR aims to identify the impacts of GPZ on green innovation and sustainability.

Although GPZ constitute a relevant and growing research topic, few studies have systematically compiled and synthesized the existing body of knowledge, and the literature presents notable contradictions across authors' findings. Accordingly, this study seeks to inform managers and policymakers by providing insights to improve existing sustainable policies and/or to support the implementation of GPZ in regions or countries seeking to replicate this model.

This article is structured in five sections. The '[Introduction](#)' presents the evolution of the research problem and discusses the topic. The subsequent section outlines the main concepts related to green finance, GPZ, and green innovation. The '[Methodology](#)' section describes the procedures and analytical path adopted for the SLR. The '[Results](#)' section discusses the main findings, and the final section presents the study's '[Conclusions](#)'.

2 Theoretical foundation

2.1 Green finance

Although there is no universally accepted definition of green finance, the term generally refers to the integration of environmental considerations into financial decision-making [2, 20, 31, 32].

According to the *International Finance Corporation* [33], an agency of the World Bank, green finance refers to the provision of financial resources that generate environmental benefits. The United Nations Framework Convention on Climate Change (UNFCCC) uses the term climate finance to refer to local, national, or transnational financing from public, private, or alternative sources to support mitigation and adaptation actions related to climate change. Zhang et al. [31] argue that the core of both concepts lies in the use of financial instruments to address climate change and broader sustainability challenges.

In contrast, China defines green finance in its "Guidelines for the Establishment of a Green Financial System" as financial activities that provide specialized services for investment, financing, and the operation and management of environmental protection projects [5, 20].

The main objectives of green finance include contributing to sustainable development and environmental protection [34], reducing greenhouse gas emissions and mitigating global warming [23, 27, 35], optimizing the allocation of financial resources [32, 36], and promoting green technological innovation [4, 37].

To achieve these objectives, green finance employs a range of instruments, including green credit [13], green bonds, green funds and green insurance [27], emissions trading and other environmental markets—such as pollution, water rights, and energy rights markets [36]—as well as government incentives and policy support [23].

2.2 Pilot areas of reform and green financial innovation

GPZ were established in June 2017 across eight locations in five Chinese provinces: *Huzhou* and *Quzhou* in *Zhejiang Province*; *Ganjiang New District* in *Jiangxi*; *Guangzhou* in *Guangdong*; the New District of *Guyan* in *Guizhou*; and *Hami*, *Changji*, and *Karamay*

in *Xinjiang* Province. Subsequently, the program was expanded to the *Lanzhou* New District in *Gansu* Province in 2019 and to *Chongqing* in 2022 [19].

According to Zhang et al. [13], the GPZ policy emerged as a response to mounting resource constraints and environmental pressures that compelled the Chinese government to pursue a green transformation of economic development, acknowledging the limited effectiveness of traditional environmental protection measures. Additionally, the policy reflected the need to strengthen financial mechanisms for green transformation and to address shortcomings of the conventional financial system.

The primary objective of the GPZ policy is to establish a green financial services system that integrates environmental protection principles into financing and investment activities. This policy promotes allocating financial resources to environmental protection and energy conservation, as well as to green industries, thereby supporting sustainable development and climate change mitigation [6, 11, 14, 20, 21, 23, 24].

Sun et al. [38] argue that this policy constitutes a comprehensive, multifaceted green finance framework, integrating multiple market-based environmental instruments and regulatory measures [6, 28].

Each green pilot zone was designed to develop a green financial system tailored to its specific priorities, local characteristics, and comparative advantages [11, 30, 39]. The policy follows a "bottom-up" approach, emphasizing experimentation and accumulation of localized experiences that can later be scaled up nationwide, in line with Deng Xiaoping's reform model [13, 23, 34].

GPZ achieve their objectives through a combination of measures, including support for financial institutions; the development of traditional green financial instruments, such as green credit and bonds, and the introduction of innovative tools, including environmental capital mortgages, environmental insurance and green investment funds; the establishment of markets for environmental rights (pollution, water and energy) and environmental credits; as well as government support and risk prevention mechanisms [27, 28, 36].

2.3 Green innovation

The concept of green innovation, or green technological innovation, was first introduced in 1994 by Braun and Wild [2, 38]. Subsequently, in 2000, green innovation was defined as the development of new technologies, products, services, and processes aimed at environmental protection [2, 27]. According to Dong et al. and Irfan et al. [2, 27], this definition has been widely accepted, as it explicitly links green innovation to the broader framework of sustainable development, encompassing technological, social, and institutional innovation [40].

Green innovation encompasses technologies, processes, and products that reduce environmental pollution and energy and raw material consumption [2, 38]. Its objective is to foster a cleaner ecological environment and more efficient economic growth by enhancing resource efficiency, mitigating climate change, and reducing greenhouse gas emissions [35].

Scholars in this field frequently use patent data as a proxy for green innovation, given its capacity to capture both the output and dynamism of innovative activity [2, 4, 16, 41].



Fig. 1 Stages of the SLR process (Adapted from Filgueiras and Melo [46])

Based on the aforementioned concept, the literature distinguishes several categories of green innovation patents, including:

- Invention Patents are generally regarded as high-quality or substantive innovations, as they represent novel and original technical solutions that advance technological development, enhance competitive advantage, and contribute to environmental protection [1, 2, 23, 27, 41];
- Utility Model Patents represent strategic innovations primarily aimed at addressing regulatory or market demands related to the quantity and speed of innovation. They are generally considered less technologically novel than invention patents [1, 4, 13, 27];
- Design patents are generally regarded as the lowest tier in terms of technological innovation, as they focus primarily on aesthetic or ornamental features rather than functional or technical advancements [4].

Although some researchers regard patents as the most appropriate measure of innovation output [30, 40, 41], other studies employ research and development (R&D) outputs for this purpose [25].

In summary, green innovation is a key driver of the transition toward a sustainable, low-carbon economy. Moreover, understanding the complex mechanisms through which GPZ shape and promote green innovation is essential for its effective advancement and diffusion.

3 Methodology

This study employed a SLR to identify and synthesize the impacts of pilot zones on the green finance reform and innovation. According to Morandi and Camargo [42], the SLR is a rigorous, structured, and transparent research methodology designed to identify, evaluate, and synthesize available evidence on a given research problem, thereby providing a comprehensive and reliable understanding of the topic [43]. The method follows well-defined protocols and emphasizes reproducibility by clearly specifying article selection procedures, inclusion and exclusion criteria, and the analytical process adopted [44, 45]. The SLR was conducted in accordance with the framework proposed by Filgueiras and Melo [46], which comprises three stages: planning, conducting, and reporting the review (see Fig. 1).

Table 1 Keyword combinations used in the SLR

Thematic	Connectors	Description of the topics
Innovation	And	Green finance reform and innovation pilot zone
Innovation	And	Pilot zones for green finance reform and innovation
Innovation	And	PZGFRI
Innovation	And	GFRI

During the planning phase, the researcher and advisors discussed relevant scientific literature to identify potential analytical dimensions of the study. On this basis, the research problem, keywords, and databases were defined. The selected databases were the *Web of Science (WoS) – Main Collection (Clarivate Analytics)* – a widely recognized source with substantial scientific coverage that underpins the *Journal of Citation Reports (JCR)* [31, 47]; and *Scopus*, which is similarly recognized for the quality and breadth of its indexed journals and articles [48]. Subsequently, multiple keyword-Boolean operator combinations were tested, resulting in the search strategy presented in Table 1.

Several single keywords related to the theme—such as “*pilots zones*”, “*green finance*”, and “*environmental regulation*”—were initially tested. These searches yielded a large number of publications in both the Web of Science and Scopus databases; however, the vast majority were not directly related to China’s GPZ. Consequently, the final set of keywords was selected based on their usage in prior relevant studies, as presented in Table 1.

The SLR was conducted exclusively in English, as it is the predominant language of scientific communication [46]. Accordingly, searches were conducted in the databases between April 10 and April 16, 2025, using the “Topic” field in Web of Science and the “TITLE-ABS-KEY” field in Scopus. Only peer-reviewed scientific articles were considered. No restrictions were imposed on the publication period due to the topic’s contemporary nature; however, as China’s GPZ were established in 2017, the resulting literature comprised studies published from 2022 onward. The search *strings* applied are presented in Table 2.

Table 3 presents the inclusion and exclusion criteria applied in the SLR, along with the rationale for the exclusion criteria.

The articles included in the SLR were selected based on the aforementioned inclusion and exclusion criteria. Figure 2 illustrates the article selection and filtering process.

In April 2025, 255 records were retrieved from the Web of Science and Scopus databases without applying any exclusion criteria. After applying the first criterion (peer-reviewed scientific articles), the sample was reduced to 247 records. The application of the second and third criteria (English-language publications and duplicate removal) further reduced the dataset to 131 articles. Duplicate records were identified and removed using RStudio.

Subsequently, upon application of the fourth criterion (screening of titles and abstracts), 71 articles were excluded based on title analysis, and an additional 12 articles

Table 2 Search string schema used in the SLR

Database	String schema
WOS—Web of Science	Green finance reform and innovation pilot zone (topic) AND Innovation (topic) OR pilot zones for green finance reform and innovation (topic) AND Innovation (topic) OR PZGFRI (topic) AND Innovation (topic) OR GFRI (topic) AND Innovation (topic) AND English (Idiomatic expressions) AND Journal articles (document types)
Scopus	(TITLE-ABS-KEY (Green finance reform and innovation pilot zone) AND TITLE-ABS-KEY (Innovation) OR TITLE-ABS-KEY (pilot zones for green finance reform and innovation) AND TITLE-ABS-KEY (Innovation) OR TITLE-ABS-KEY (PZGFRI) AND TITLE-ABS-KEY (Innovation) OR TITLE-ABS-KEY (GFRI) AND TITLE-ABS-KEY (Innovation)) AND LIMIT-A (LANGUAGE, "English") AND LIMIT-A (DOCTYPE, "one"))

Table 3 Inclusion and exclusion criteria applied in the SLR with justification for exclusions

Inclusion criteria	Exclusion criteria	Justification
CI 1 Peer-reviewed scientific articles	CE 1 Publications other than peer-reviewed scientific articles	Works other than peer-reviewed scientific articles were considered outside the scope of this review, as peer-reviewed publications undergo formal evaluation by subject-matter experts and represent validated scientific contributions
CI 2 Articles in English	CE 2 Non-English-language publications	Given the central role of English in scientific communication, studies published in languages other than English were excluded from the review
CI 3 Unique records identified across the selected databases	CE 3 Duplicate records across databases	To avoid redundancy and ensure each study was counted only once, duplicated records indexed simultaneously in Web of Science and Scopus were removed
CI 4 Articles related to GPZ	CE 4 Articles not directly addressing the topic, as evidenced by the title or abstract	Following screening of titles and abstracts, studies not directly related to GPZ and innovation were deemed irrelevant to this review
CI 5 Full-text reading of the articles	CE 5 Articles with paid or unavailable full-text access	Articles that were unavailable or accessible only through paywalls were considered outside the scope of this study because their full content could not be accessed

were excluded following abstract review, resulting in 48 remaining studies. Finally, the fifth criterion (full-text assessment and exclusion of paid or inaccessible articles) was applied, yielding a final sample of 47 articles that met all established criteria.

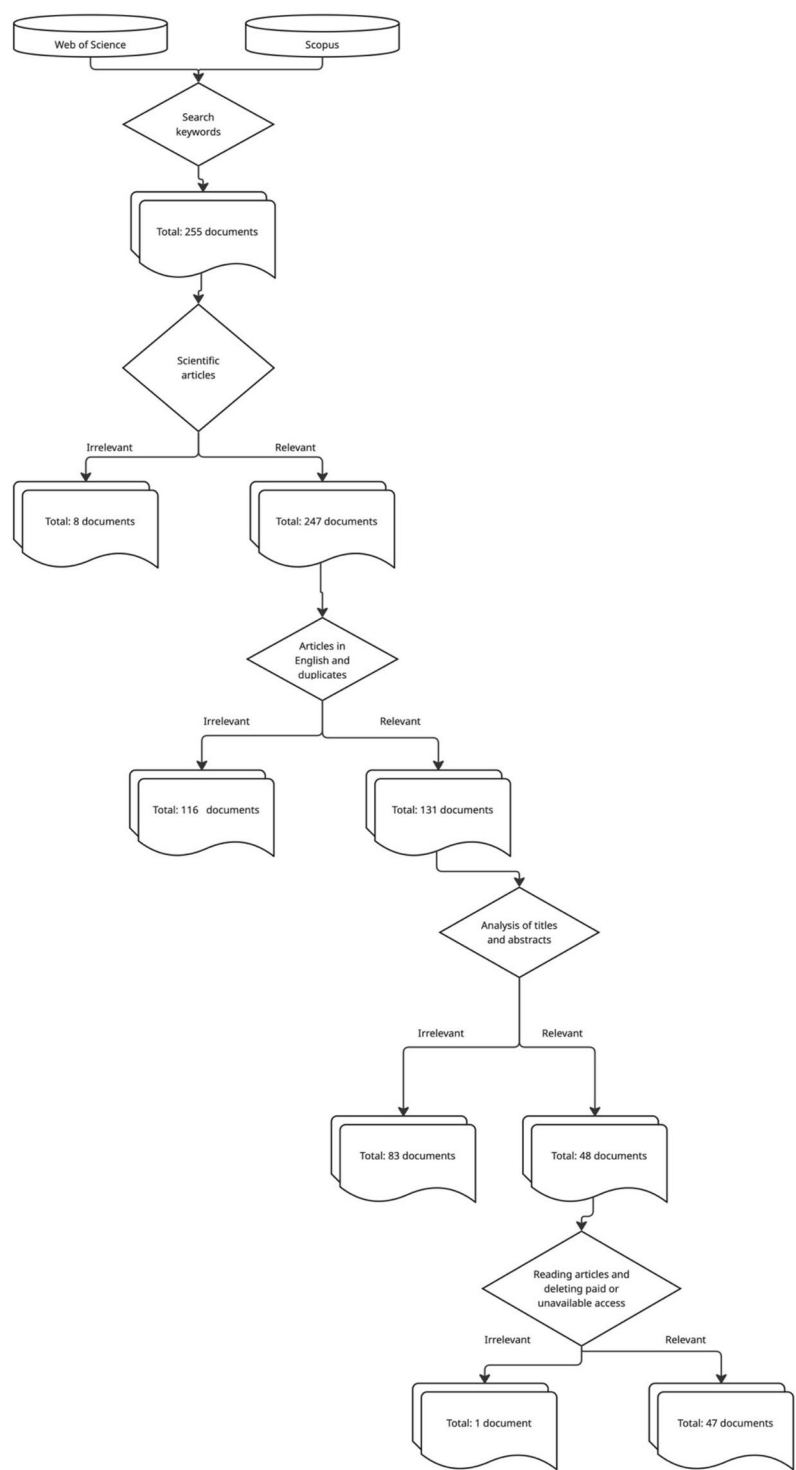


Fig. 2 Selection and filtering process based on the inclusion and exclusion criteria

Screening based on the fourth (screening of titles and abstracts) and fifth (full-text assessment) criteria was conducted collaboratively by the authors. No disagreements arose during this process—all articles directly related to GPZ, innovation, and sustainability were included to ensure broad coverage of the existing literature.

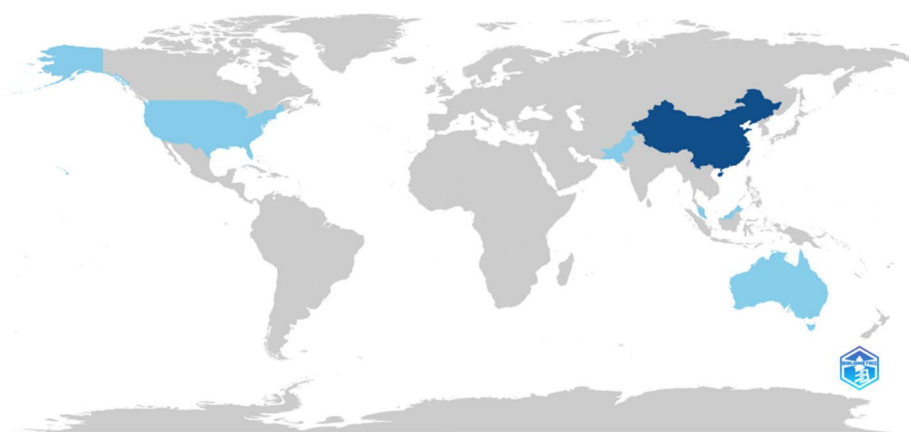


Fig. 3 The geographical distribution of publications

Nevertheless, it is important to note that the final set of articles included in this SLR represents only a specific subset of the broader literature on GPZ. Studies were intentionally restricted to those that directly linked GPZ to innovation and sustainability, and were indexed in the databases aforementioned. In addition, articles that required paid access or were otherwise inaccessible for full-text reading in these databases were excluded. Consequently, the overall body of research on GPZ is broader and may include studies that examine the policy from perspectives beyond sustainability and innovation.

Thus, a total of 47 articles remained after all exclusion criteria were applied. According to Filgueiras and Melo [46], the rigor of a SLR lies not in the number of included studies but in the careful curation of articles through predefined protocols and criteria, which ensure the transparency and reliability of the review process [49].

4 Results

4.1 Descriptive analysis

For the descriptive analysis, the R software environment [50] and the bibliometrix package [51] were employed to systematize the selected articles by year of publication, map journals by country, and identify the most influential journals and authors. Subsequently, a word cloud was generated from all keywords extracted from the articles, along with a keyword co-occurrence network.

4.2 Publication trends over time

From 2022 onward—five years after the establishment of the GPZ—scholarly publications on the topic began to emerge, with 7 articles published in 2022 alone. A consolidation phase followed in 2023, marked by an increase of more than 150% in publication output, totaling 18 articles. This upward trend continued into 2024. By early April 2025, four articles had already been published.

An analysis of publications by country revealed a strong concentration of studies originating from China. At the same time, contributions from the United States, Malaysia, Pakistan, and Australia were also identified; no publications were found from Latin America, Africa, or Europe. The geographical distribution of publications is illustrated in Fig. 3.

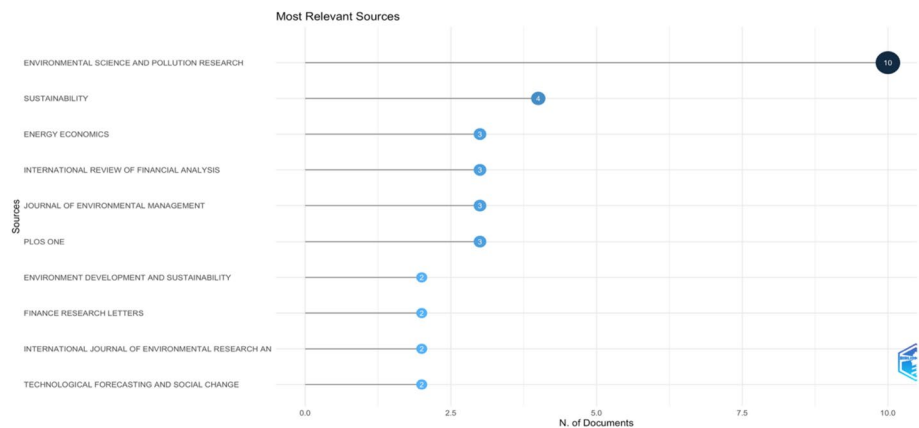


Fig. 4 The most influential journals with publications on the topic

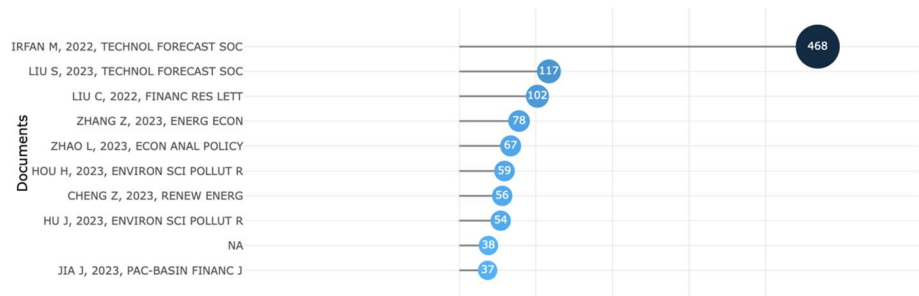


Fig. 5 Most cited publications worldwide on GPZ

The journals with the highest number of publications on the topic were Environmental Science and Pollution Research (10 articles) and Sustainability (4 articles), followed by the International Review of Financial Analysis, Journal of Environmental Management, PLoS One, and Energy Economics (3 publications each), as illustrated in Fig. 4.

Figure 5 displays the most highly cited studies at the global level, highlighting the studies of Irfan et al. [2] (468 citations), Liu et al. [34] (117 citations), and Liu et al. [21] (102 citations). Figure 6 presents the most prolific authors in the field, with Zhang Z standing

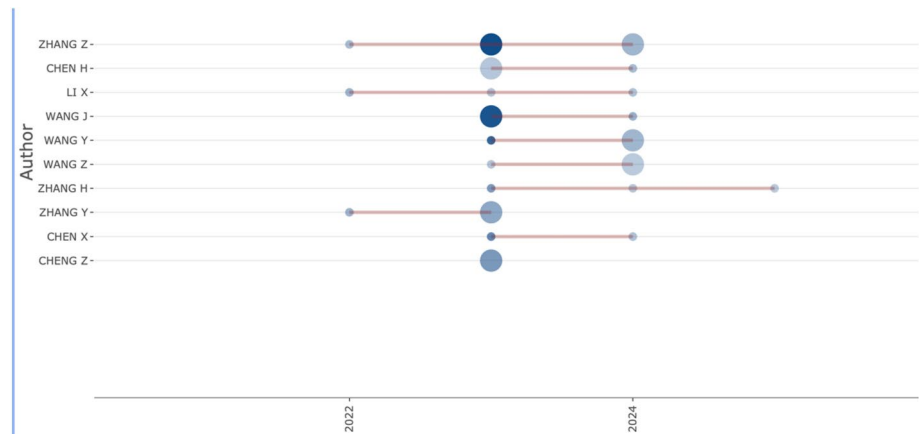


Fig. 6 Authors' production output on GPZ



Figure 7 presents a word cloud generated from the keywords of all included articles, in which larger font sizes and more intense colors indicate higher frequency of use. Among the most prominent terms related to the theme are the expressions green finance, green innovation, green technological innovation, DiD model, green financial policy, and China. Additional terms with lower frequencies include green patents and sustainable development.

Finally, a keyword co-occurrence analysis was conducted. A total of 36 terms appeared at least twice across the selected studies. These keywords were organized into five main *clusters* and visualized by frequency of occurrence (node size) and cluster membership (color), as shown in Fig. 8.

Cluster 1 (red) comprised 12 terms: green finance, China, growth, sustainable development, green patents, consumption, DiD model, enterprise, green economy, management, pollution, and social responsibility. *Cluster 2* (blue) included 11 terms: green innovation, credit, impact, performance, mechanism, productivity, innovation, investment, policy, economic growth, and moderating effect. Cluster 3 (green) consisted of two terms: the DiD model, and the green finance pilot policy. Cluster 4 (purple) comprised eight terms, including credit policy, green technological innovation, and green finance policy, constraints, economy, enterprises, political effect, and polluting enterprises. Finally, *cluster 5* (yellow) comprised three terms: environmental regulation, difference model in differences, and *regulations*.

4.3 Narrative analysis: main trends

Based on the analysis of the 47 selected articles, the main trends regarding the impact of GPZ policies on green innovation were identified. The synthesized findings are summarized in Table 4.

Regarding the geographical location, the impact of GPZ on green innovation and green total factor productivity is most pronounced in China's eastern region [18, 27, 30, 40]. Among the 14 studies that explicitly examine regional heterogeneity, approximately 70% identify the eastern region as exhibiting the strongest policy effects. This stronger effect is likely driven by higher levels of economic development, more advanced infrastructure, greater concentration of resources, a higher human capital endowment, and the presence of more mature companies, which together favor research and development and green innovation [6, 19, 41].

In 90% of studies on market characteristics, the impact of GPZ was more pronounced in more competitive markets, characterized by stronger market discipline, greater transparency, and more efficient resource allocation within a fairer financial system. Such environments enhance the effectiveness of green financial policies, encouraging companies to pursue green innovation as a source of competitive advantage [6, 23, 25, 28].

According to all studies on GPZ versus innovation [4, 6, 16, 24–26, 28, 30, 32, 38, 40], more developed localities and cities, characterized by higher GDP *per capita*, also exhibit stronger innovation responses to the implementation of GPZ.

The impact of GPZ is also stronger in non-resource-based cities, as they typically exhibit economic structures that are less dependent on natural resource-extractive industries, which are generally more polluting and energy-intensive [15, 16, 19, 24, 55].

Regarding environmental regulation, Zhang et al., Tang et al., Lin et al., Liu et al., Wang et al., Lin et al., Huang et al., and Zhang et al. [4, 15, 16, 23, 28, 30, 39, 52] argue that the effects of GPZ policies are more pronounced in regions with stricter regulatory frameworks. This characteristic is reported in 88% of the studies that include environmental regulation as a variable in their empirical analyses.

The literature does not converge on a dominant pattern concerning company ownership. Approximately 40% of the studies reported stronger effects on state-owned enterprises [3, 18, 23, 27, 32, 37], while about 60% suggested greater benefits for private companies [12–14, 20, 28, 29, 34, 41, 54], highlighting substantial heterogeneity in policy's outcomes.

Table 4 Main trends about pilot areas and green innovation

	Characteristics	Authors
Geographical location	The impact on green innovation and green productivity was most pronounced in eastern China	Mao and Chen, Tang et al., Dong et al., Zhang and Li, Ge et al., Hou and Shi, Lu et al. [6, 18, 19, 27, 30, 40, 41]
Market competitiveness	Regions with more competitive markets were more strongly affected by GPZ policies	Huang et al., Chen et al., Ge et al., Xie [6, 24, 25, 28]
Development level of localities	More developed localities and cities positively affect green innovation	Sun et al., Mao and Chen, Liu, et al., Huang et al., Ge et al., Du et al., Xie et al., Zhang et al., Chen et al., Yang et al. [4, 6, 16, 24–26, 28, 30, 32, 38, 40]
Resource heterogeneity	The policy's impact was most pronounced in non-resource-based cities (extractive cities)	Chen et al., Hou and Shi, Wang et al., Zhang et al. [15, 16, 19, 24]
Environmental regulation	Locations with stricter environmental regulations experienced a stronger positive impact from the policy	Zhang et al., Tang et al., Lin et al., Liu et al., Wang et al., Lin et al., Huang et al., Zhang et al., Cui et al. [4, 15, 16, 23, 28, 30, 39, 52, 53]
Firm ownership	The literature presents contradictory evidence regarding firm ownership. While several studies suggest that state-owned enterprises benefit more from the policy, others find that private companies experience a greater impact on green innovation. As a result, no clear ownership-based trend can be identified	Zhang and Li, Zhang and Wei, Yang et al., Dong et al., Lin et al., Zeng et al., Liu and Xiong, [3, 18, 21, 23, 27, 32, 37] Lu et al., Liu and Wang; Yang et al., Zhang, W., et al., Lyu et al., Huang et al., Jia et al., Zhao et al., Dong and Tao [12–14, 20, 28, 29, 34, 41, 54]
Firm size	Larger companies tend to experience a stronger positive impact from GPZ	Xie et al., Chen et al., Zeng et al., Dong et al., Huang et al. [24, 25, 27, 28, 37, 38]
Pollution intensity	The GPZ policy exerts a chilling effect on green innovation and total factor productivity among highly polluting companies	Zhang et al., Zeng et al., Liu et al., Lin et al., Ran and Zhang, Zhang and Dong [1, 4, 13, 17, 23, 37]
Access to credit	Green innovation is promoted through alleviating financing constraints, reducing costs, and increasing the share of long-term financing At the same time, the policy induces a redistribution of capital, reallocating financial resources from highly polluting companies to greener ones. This reallocation raises financing costs and tightens credit constraints for polluters, restricting non-green investments and compelling companies to undertake green transformation	Zeng et al., Lu et al., Fu et al., Wang et al., Ge et al., Dong et al., Liu and Xiong [6, 21, 27, 35, 37, 41, 55] Zhou et al., Yang et al., Zhao et al., Huang et al., Jia et al., Li et al. [12, 14, 22, 28, 29, 56]
Digitization	GPZ contribute to promoting green innovation and sustained innovation, while enhancing capital allocation and efficiency in the green transformation of enterprises	(Hou et al., Liu et al., Liu et al. [4, 5, 57]

Regarding firm size, 80% of studies addressing this dimension consistently reported that larger firms experience stronger positive effects from implementing GPZ [24, 25, 27, 28, 37, 38].

Regarding pollution intensity, the GPZ policy exerts an inhibiting effect on highly polluting companies, constraining both the quality of their green innovation [1, 4, 13, 23, 37] and their total factor productivity [14, 32]. Moreover, Zhang and Dong [17]

identified a polarization effect, in which the policy promotes green innovation among environmentally friendly companies while simultaneously suppressing innovation in polluting enterprises. Only 15% of the studies reported a positive impact of the policy on highly polluting companies.

Access to credit represents one of the central mechanisms through which GPZ influence innovation outcomes. The policy stimulates green innovation by easing credit constraints for environmentally sustainable companies [27, 37, 41]. Simultaneously, it raises financing costs for polluting enterprises, thereby exerting pressure for technological upgrading and structural transformation [22, 23, 28]. This mechanism can be interpreted as a reallocation of capital flows, increasing the availability of financial resources for green companies while restricting access for polluting companies through higher financing costs. This trend is reported by 87% of the studies.

Another important mechanism through which GPZ promote innovation is digitalization [4, 5]. Digital transformation enhances the persistence of innovation by strengthening companies' continuous innovation capacity, while also improving capital allocation efficiency during enterprises' green transformation [57].

5 Discussion

The descriptive analysis revealed an overwhelming predominance of studies conducted in China, which is expected given the policy's national scope. However, the scarcity of research from other developing countries is noteworthy, given that GPZ represent an innovative policy for achieving sustainable development goals and have the potential to serve as a transferable model for other countries, provided that local challenges and institutional particularities are taken into account [2, 4, 17, 26, 30, 58].

Regarding the methodologies used to evaluate the effects of implementing the GPZ policy, 87% of the studies employed the DiD model. In 78% of these cases, the standard DiD specification was applied, corresponding to 37 articles. Variations of the DiD approach were used as follows: Difference-in-Difference-in-Difference (DDD) in 8.51% of the studies (four articles); staggered DiD in 4.21% (2 studies); and PSM-DiD, generalized DiD, time-varying DiD, and Difference-in-Synthetic-Differences (DiSD), each accounting for 2.12% and appearing in one article each. According to Lu et al. and Zhang et al. [39, 41], DiD is the most effective econometric model for evaluating public policies. Other methodologies include the Synthetic Control Method (SCM), used in 6.38% of the studies (3 articles), and the Generalized Synthetic Control Method (GSCM), applied in 2.12% of the studies (1 study) [6, 24, 30, 40]. In addition, game theory combined with DiD [59] and baseline model combined with DiD and PVAR [2] were each employed in one study (2.12%). Figure 9 illustrates the distribution of econometric models used across the 47 reviewed articles.

Regarding the methodologies employed to assess the effects of GPZ implementation, approximately 87% of the studies adopted the DiD approach or its extensions, which is widely regarded as one of the most effective econometric methods for evaluating public policies [39, 41]. Other methodological approaches include the Synthetic Control Method [6, 24, 30, 40] and Game Theory-based models [59].

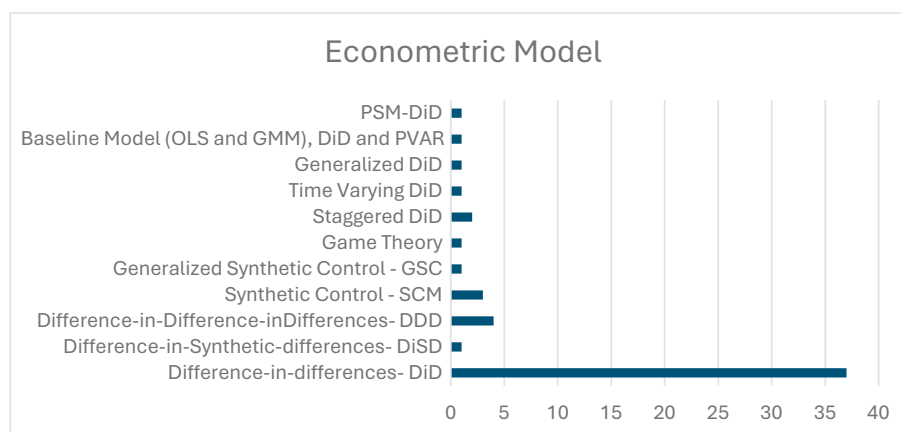


Fig. 9 Econometric models

Despite the identification of dominant methodological choices and general empirical trends, the narrative analysis revealed substantial divergences among the studies, as discussed below.

Although most studies have identified stronger effects of GPZ in China's eastern region, some evidence suggests more pronounced effects on green total factor productivity in western regions [4], possibly reflecting greater demand for green financial instruments in these less developed areas [1, 15].

While most studies reported stronger effects in highly competitive markets, Zhou et al. [22] showed that GPZ have a greater impact on green export sophistication in regions with lower market development, indicating that credit pressure may also induce company modernization.

Some studies have reported stronger effects of GPZ in smaller cities [15, 35, 55, 60], which diverges from the majority of the literature, which finds more pronounced positive impacts in larger urban centers.

In contrast to the aforementioned studies, Zhang and Li [18] reported divergent findings from an analysis of micro-enterprises, showing that the GPZ policy had no significant effect on the number of green innovations in companies located in the central and western regions, which are traditionally non-resource-based areas.

Although strict environmental regulations are widely regarded as a cornerstone of green innovation, Zhang et al., Zhang and Dong, and Ran and Zhang [1, 11, 17] find that, under certain conditions, the effects of the GPZ policy are more pronounced in regions with relatively more flexible environmental regulations.

The greatest divergence in the literature emerges when company ownership is considered, particularly the distinction between state-owned and private enterprises. Some studies report that green zones exert a stronger impact on state-owned enterprises, largely because these companies can respond more rapidly to national policy directives and typically face fewer financing constraints [3, 18, 23, 27, 32, 37]. In contrast, other studies present evidence to the contrary, indicating that policy effects are more pronounced among private companies [12, 13, 28, 29, 34, 41, 54]. Regarding total factor productivity, Zhao et al. and Dong and Tao [14, 20] find that private enterprises benefit more significantly from the policy.

In contrast to studies emphasizing stronger policy effects among large companies, Zhao et al. and Ran and Zhang [1, 14] identified an inhibitory effect of GPZ on large enterprises. This perspective is reinforced by Yang et al., Du et al., and Cheng et al. [12, 26, 61], who found that the policy disproportionately benefits small and medium-sized enterprises in the renewable energy sector. Additionally, Zhao et al. [14] reported that GPZ encourage greater environmental investment among small and medium-sized enterprises.

Contrary to most findings in the literature, Chen et al., Lin et al. [23, 24, 62] reported a positive effect of GPZ on green innovation in highly polluting companies. However, when considering access to credit as a transmission mechanism, Ran and Zhang and Zhang et al. [1, 13] identified a penalty effect of the policy, leading to a reduction in green innovation among polluting enterprises rather than facilitating their technological transformation.

Despite the contradictions and divergences identified by this SLR, all studies converged in indicating that GPZ policy plays a significant role in promoting environmental sustainability and green development, although they may generate adverse or limited effects in certain contexts.

6 Conclusion

China's GPZ policy was introduced in 2017 to foster green finance, promote coordinated economic and environmental development, and stimulate innovation toward a low-carbon economy. After five years of implementation, a growing body of academic research has examined the policy's effects from multiple perspectives in the regions where it was applied. This study is likely among the first to systematically compile and synthesize the existing literature on the policy's impacts, reporting the main findings and highlighting the divergences identified across studies.

The rise in global temperatures and the increasing frequency of extreme and catastrophic weather events underscore the urgency of effective actions to address climate change. This study analyses the impacts of the GPZ policy. It synthesizes existing knowledge on the subject, offering insights that may support policymakers and public authorities in improving current sustainability policies and in adapting or implementing the GPZ framework in other regions or countries seeking to replicate this model.

Additionally, this study sought to identify how the GPZ policy affects localities; however, substantial contradictions emerge across the literature. Divergent findings are observed across geographical location, regional level of development, company size, pollution intensity, ownership structure, environmental regulation, and access to credit and financing.

The theoretical contribution of this study lies in identifying the key variables affected by the GPZ policy and synthesizing how the literature on this topic has evolved, without attempting an in-depth analysis of each dimension. By doing so, this review highlights areas of convergence and, more importantly, variables for which the evidence remains contradictory.

In this sense, the contradictions identified in this study suggest the need for further research to examine the specific mechanisms by which GPZ policies can positively impact public and private enterprises of varying sizes, as well as cities at different stages

of development. Combined with the reported trends showing a lower degree of contradiction, such studies could provide policymakers with more assertive pathways.

Significant future research could indicate to what extent stricter environmental legislation, which limits access to credit and imposes financing restrictions on polluting companies, can stimulate green innovation rather than generating an inhibitory effect, as some studies suggest.

Furthermore, this study indicates to policymakers that the effectiveness of pilot zones does not stem from a generic application. In policy design, it is essential to evaluate the distinct characteristics of regions and adapt interventions to achieve better results. Regions with varying levels of development, differing market maturity, and distinct levels of financial and banking sector advancement require tailored interventions to be successful.

It is important to emphasize that this review reports findings on GPZ as described in the original articles, without using formal tools to assess methodological quality or to systematize the robustness of these papers. As this is an exploratory study, we chose to report all articles that passed the screening process, excluding those that did not. Another limitation of this study is the intentional use of restrictive search criteria, which directly link GPZ to innovation and sustainability. Consequently, the reported findings refer only to a subset of the literature on GPZ, excluding relevant studies that analyzed the theme from other perspectives.

It should also be noted that, considering China's characteristics such as a single central government promoting a 'top-down' model with state intervention in resource allocation and state-owned enterprises, the structure of its financial sector, and its stage of development compared to developed countries that have already addressed environmental issues China is still seeking to resolve, the findings in this study may serve as a compass for developing regions, particularly for the Global South. However, in other contexts, such as Western developed nations, further studies in those regions would be necessary for the trends presented here to be universally accepted.

Finally, the findings underscore the complexity and multifaceted nature of the transition toward sustainability. While GPZ exert a strong catalytic effect, their impacts are heterogeneous, highlighting the need for policymakers to account for contextual differences during policy design and implementation.

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Author contributions

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